SHARDA UNIVERSITY — SCHOOL OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING | CSCR2600 PBL-1

Project Synopsis & Proposal: Expenzo

A Verifiable Web Platform for Personal Finance Intelligence & Deterministic Budget Monitoring

Course Milestone: CSCR2600 Project-Based Learning (PBL) - 1 | Rubric #R0 (Zeroth Evaluation)

Academic Session: 2026–2027 | **Program:** B.Tech CSE (AI & ML)

Project Title: Expenzo — Personal Financial Intelligence Platform

Confirmed Team: Aryan Kaundilya (Architecture & Backend); Devanshi Singh (Frontend UI/UX & QA)

1. Clarity of Problem Definition (Rubric R0 - PO1)

Managing personal finances is a foundational life skill, yet young adults and students struggle with manual tracking, floating-point rounding errors, privacy risks, and hallucinatory financial advice. Expenzo provides an integer-precise, user-isolated personal finance platform combining relational PostgreSQL storage, reactive budget alerts, and verified dual-layer financial intelligence.

2. Description of Project Concept & Approved Scope (Rubric R0 - PO4)

Expenzo integrates double-entry accounting principles with an intuitive web application. In-scope capabilities include multi-account depository tracking, canonical transaction logging, atomic internal transfers, category budgeting with 80%/100% threshold alerts, and grounded financial inquiry handling. Excluded scope includes direct bank credential scraping, real fund disbursement, and automated speculative stock trading.

3. Study of the Literature Survey (Rubric R0 - PO5)

Expenzo grounds its design in validated behavioral and software engineering literature:

Source	Core Finding	Expenzo Application
Thaler (1985) *Mental Accounting*	Categorization dictates willingness to spend.	Informs category budget envelopes and threshold alerts.
Soman (2001) *Payment Mechanisms*	Abstract payment decreases pain of paying.	Motivates instant ledger updates and balance reconciliation.
Prelec & Loewenstein (1998)	Consumption decoupling causes debt accumulation.	Double-entry internal transfer ledger and liability tracking.
OWASP Foundation (2026)	Financial security requires database-level enforcement.	PostgreSQL Row Level Security (RLS) on all user data.
Google Gemini (2024)	Factuality requires strict database grounding.	Deterministic local rules + server-side Gemini Edge Function.

4. Project Feasibility Analysis (Rubric R0 - PO2)

- **Technical:** React 19 SPA, TypeScript 5.7, PostgreSQL 15 via Supabase, Vitest 2.1 automated test runner.
- **Operational:** Zero installation overhead, runs on any modern browser, privacy-preserving client design.
- **Schedule:** 4 distinct phases strictly mapped to evaluation rubrics R0 through R3.
- **Cost:** Developed on legitimate free developer tiers (GitHub, Vercel, Supabase, Google AI Studio).

5. Project Roadmap & Team Work Distribution (Rubric R0 - PO11)

Student	Role	Key Responsibilities
Aryan Kaundilya	Architecture & Backend Lead	PostgreSQL schemas, RLS policies, atomic RPCs, state store, 184 automated tests.
Devanshi Singh	Frontend UI/UX & QA Lead	Dashboard layout, Recharts data visualizers, modal dialogues, defect verification.

6. Summary of Administrative Status

This document satisfies all criteria for CSCR2600 Rubric #R0. Administrative placeholders for official university System IDs, Roll Numbers, Batch Section, and Faculty Guide are consolidated into the master Readiness Report checklist.